Peer Review Rubrics

Code Review 1 and Code Review 2

Java Desktop Application using Swing UI and JDBC

This document provides one common manual-review process for internal reviewers from the same team and external reviewers from a faculty-assigned team. Peer scores and findings support the review; the faculty records the final milestone mark.

1 Review process and mandatory records

Manual review rule: Automated code-analysis, code-quality scoring and code-review tools shall not be used. Reviewers shall read the source code, compile and run the project, manually test the user interface and database operations, and discuss the code with its authors.

1.1 Two-stage peer review

1.2 HackerRank Java progress

Mandatory in both code-review submissions and PPTs. For Code Review 1, use Milestone 3 as the previous review. For Code Review 2, use Code Review 1 as the previous review.

1.3 Evidence to submit

- The exact source-code version reviewed and instructions to run it.
- Completed internal and external rubrics.
- A findings log containing author response and recheck status.
- Current HackerRank profile information for every member.
- A short PPT presenting progress, important code decisions, review findings and corrections.

2 Review identification and assignment

2.1 Submission details

2.2 Internal reviewer assignment

2.3 External reviewer assignment

2.4 Reviewer declaration

We reviewed the identified source version manually. We did not use automated code-analysis or code-quality scoring tools. Our scores and findings are based on code reading, execution, manual tests and discussion with the author team.

Internal reviewer signature(s): [_____] External reviewer signature(s): [_____]

3 Common rating guidance

3.1 Scoring rules

- Award scores in sensible increments such as 0.25 marks where required.
- Every reduced score must have a short evidence-based comment.
- A feature that cannot be run may still receive limited design or readability credit, but not full correctness credit.
- Do not reward an advanced feature merely because it exists; it must be appropriate, understandable and integrated.
- Peer scores are recommendations. The faculty validates the evidence and records the final mark.

3.2 Finding importance

4 Code Review 1 rubric

Milestone 4 | Basic Java and OOP with focus on coding standards | Maximum 5 marks

4.1 Criterion guidance

4.2 Score summary

5 Code Review 1 manual checklist

Running and behaviour

[ ] Yes [ ] Partly [ ] No Project compiles and starts using the submitted instructions.
[ ] Yes [ ] Partly [ ] No At least one meaningful workflow operates from Swing input to a visible or stored result.
[ ] Yes [ ] Partly [ ] No Normal input produces the expected output.
[ ] Yes [ ] Partly [ ] No Invalid input does not cause an uncontrolled crash.

Basic Java

[ ] Yes [ ] Partly [ ] No Variables and data types are suitable for the stored values.
[ ] Yes [ ] Partly [ ] No Conditions and loops are correct and not unnecessarily complicated.
[ ] Yes [ ] Partly [ ] No Methods have meaningful names, parameters and return values.
[ ] Yes [ ] Partly [ ] No Repeated fixed values are replaced by named constants where appropriate.

OOP and structure

[ ] Yes [ ] Partly [ ] No Fields are private unless another access level is justified.
[ ] Yes [ ] Partly [ ] No Constructors create valid objects.
[ ] Yes [ ] Partly [ ] No Classes have understandable responsibilities.
[ ] Yes [ ] Partly [ ] No Inheritance, overriding or interfaces are used only when the relationship is meaningful.
[ ] Yes [ ] Partly [ ] No Swing, business logic and JDBC responsibilities are reasonably separated.

Coding standards

[ ] Yes [ ] Partly [ ] No Class, method, variable, constant and package names follow the guidelines.
[ ] Yes [ ] Partly [ ] No Indentation, braces, spacing and blank lines are consistent.
[ ] Yes [ ] Partly [ ] No Unused imports, variables, debug output and commented-out code are removed.
[ ] Yes [ ] Partly [ ] No Comments explain non-obvious purpose and are not misleading.

Reviewer notes

Strong points: [_____]

Most important improvement: [_____]

6 Code Review 1 findings and decision

6.1 Recheck record

6.2 Review decision

7 Code Review 2 rubric

Milestone 5 | Advanced concepts, proper JavaDocs and improved coding style | Maximum 7 marks

7.1 Criterion guidance

7.2 Score summary

8 Code Review 2 manual checklist

Advanced Java

[ ] Yes [ ] Partly [ ] No Each advanced feature solves a real project need and can be explained by the team.
[ ] Yes [ ] Partly [ ] No Collections and generic types are suitable and type-safe.
[ ] Yes [ ] Partly [ ] No Interfaces, inheritance or abstract classes express a meaningful design relationship.
[ ] Yes [ ] Partly [ ] No Enums, custom exceptions, streams, file handling or background tasks are used only where appropriate.

Reliability and JDBC

[ ] Yes [ ] Partly [ ] No PreparedStatement is used for variable or user-supplied values.
[ ] Yes [ ] Partly [ ] No Connections, statements and result sets are closed safely.
[ ] Yes [ ] Partly [ ] No Transactions are used where related changes must remain consistent.
[ ] Yes [ ] Partly [ ] No Exceptions are not ignored and user messages are meaningful.
[ ] Yes [ ] Partly [ ] No Valid, invalid, duplicate and database-failure cases have been considered.

JavaDocs

[ ] Yes [ ] Partly [ ] No Important classes and interfaces state their responsibility.
[ ] Yes [ ] Partly [ ] No Public or protected methods have useful descriptions where required.
[ ] Yes [ ] Partly [ ] No @param, @return and @throws are complete and accurate where applicable.
[ ] Yes [ ] Partly [ ] No Documentation matches the current method signature and behaviour.
[ ] Yes [ ] Partly [ ] No Comments and JavaDocs explain purpose or contract rather than obvious syntax.

Style integration and improvement

[ ] Yes [ ] Partly [ ] No Naming, formatting and package organisation are consistent across all members' code.
[ ] Yes [ ] Partly [ ] No Important findings from Code Review 1 are fixed or have an accepted explanation.
[ ] Yes [ ] Partly [ ] No Duplicated or oversized code has been reasonably refactored.
[ ] Yes [ ] Partly [ ] No Important workflows operate from Swing input through JDBC to a visible result.

9 Code Review 2 findings and decision

9.1 Recheck record

9.2 Review decision

10 Final peer-review record

10.1 Individual contribution discussion

10.2 Review comparison

10.3 Submission confirmation

[ ] The reviewed source version is clearly identified.
[ ] Internal and external reviews were completed independently.
[ ] All scores have supporting evidence or comments.
[ ] All Must Fix findings have an author response and recheck result.
[ ] Accepted corrections are present in the submitted project.
[ ] Every member's HackerRank profile and current Java stars are included.

Author Team Lead: [_____] External Review Lead: [_____] Faculty: [_____]